SurfacePrint

Typographic Fingerprints of LLMs

Wentao Zhang, Sajad Ebrahimi, Nima Jamali, Bardia Shirsalimian,
Kelly McConvey, Jalehsadat Mahdavimoghaddam, Maksym Taranukhin,
Maura R. Grossman, Vered Shwartz, Ebrahim Bagheri, Yuntian Deng

Same word. Different apostrophe.

don't

U+0027

UTF-8: 27

don’t

U+2019

UTF-8: E2 80 99

Same words, different formatting

Markdown

```
**A simple answer**

It doesn’t change.
```

Plain text

```
A simple answer

It doesn't change.
```

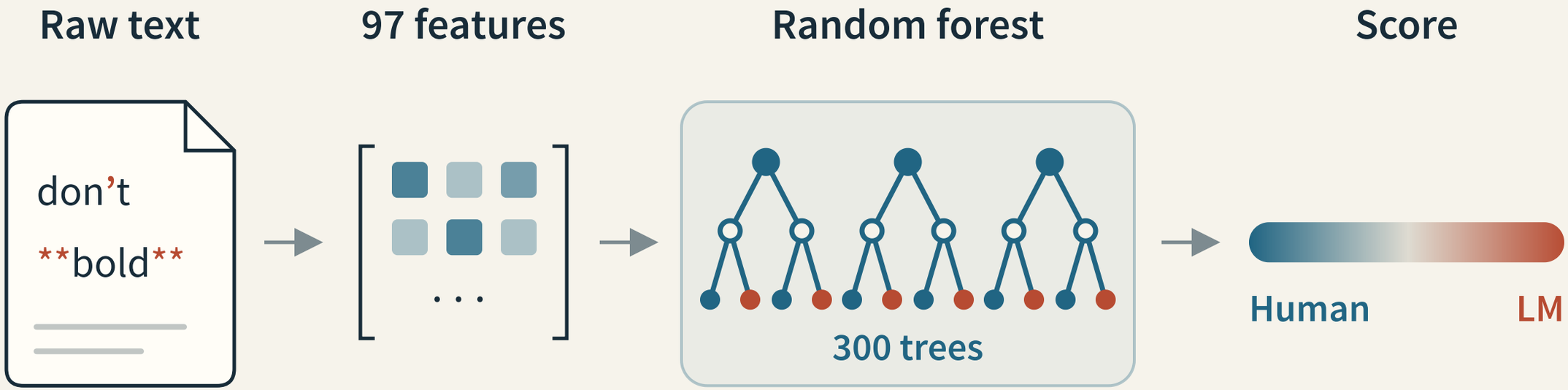
Count the formatting.



3 / 5

60% curly

A random forest learns formatting patterns.



Test new text from familiar sources.

TRAIN

Human text

+ **17 AI models**

TEST

New texts

from the same sources

Evaluation

Human text < Generated text

Detector score

0.5

Chance

AUROC

1.0

Perfect

Formatting separates these benchmark texts.

0.983

Mean AUROC

Held-out texts · 17 known model sources

Even whitespace carries a fingerprint

Raw text

```
First paragraph.


Second paragraph.
```

0.983 AUROC

Whitespace cleaned

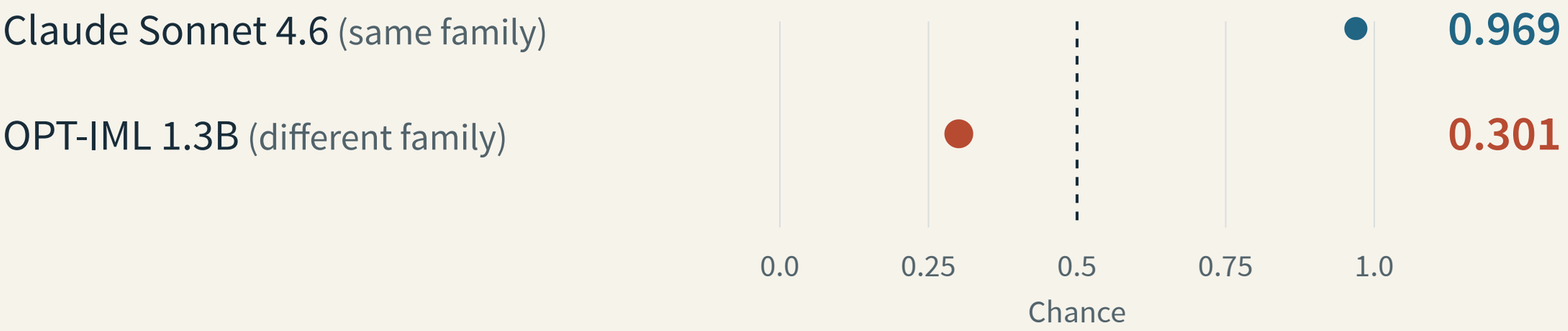
```
First paragraph.

Second paragraph.
```

0.889 AUROC

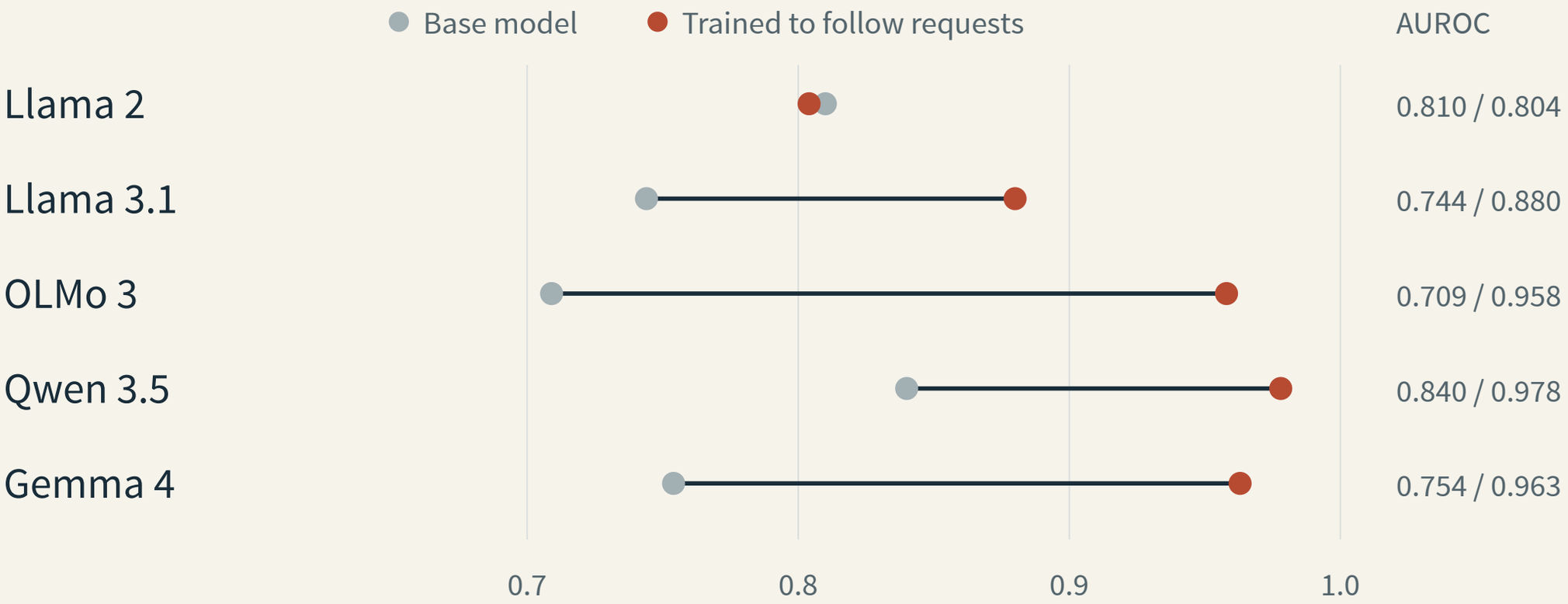
Different model families have different fingerprints

Train: human + Claude Sonnet 4 · cleaned text



Instruction-tuned text is often easier to detect

Whitespace cleaned



AI writing styles leave revealing fingerprints.

An AUROC score does not set the false-positive rate

A cutoff turns a score into a flag

Different cutoffs trade missed AI text against wrongly flagged human text

The classifier combines many small decision trees

300	6	3
trees	maximum depth	minimum leaf size

97 measurements, balanced class weights, fixed seed

One path in an illustrative tree

Depth-3 tree · AUROC 0.750

1	Markdown bold present? **Yes**
2	Double-backtick inline code present? **No**

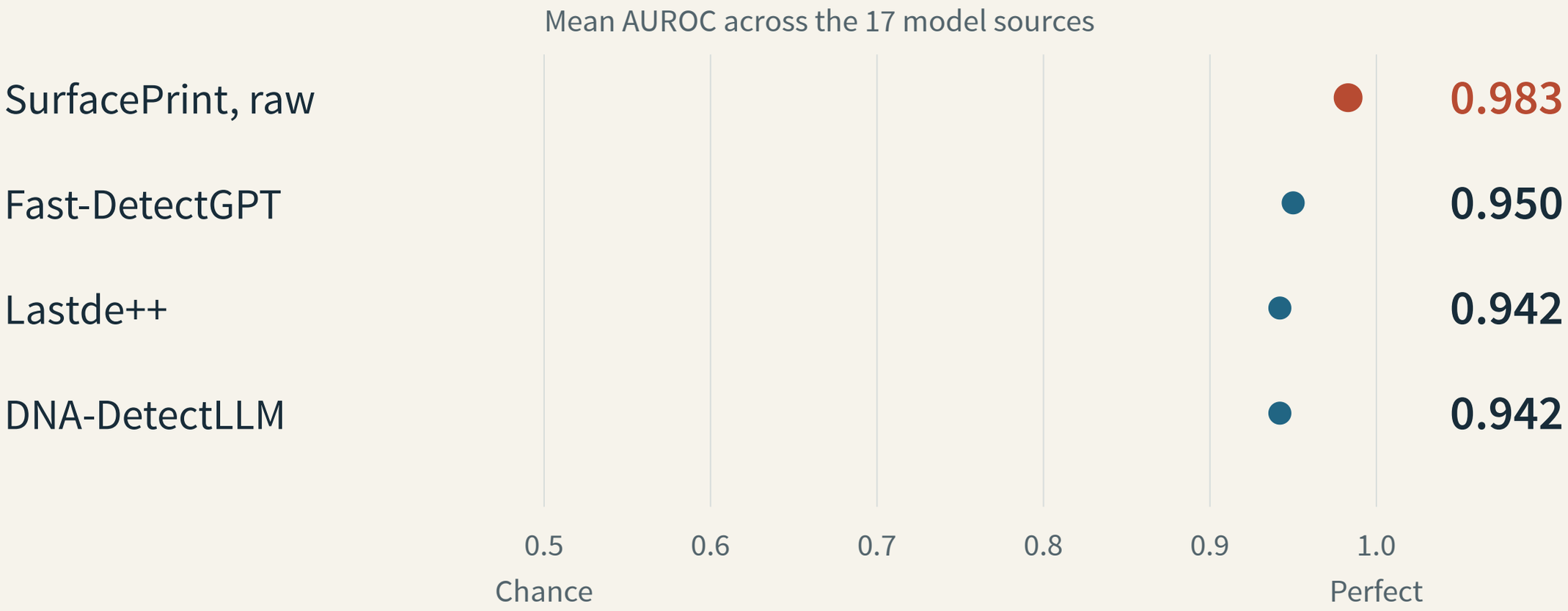
Result Tree predicts **LLM**

The 97 measurements cover five surface families

FAMILY	EXAMPLE	COUNT
Prose typography	' ’ ... …	22
Markdown	**bold** __bold__	25
Math markup	x \(x\)	24
Code formatting	Tabs spaces	20
Whitespace	Blank lines	6

The main reference-model comparators

SurfacePrint: supervised · Comparators: training-free



The source pool and detector sample differ

Source pool, per domain

2,000 train / 500 validation / 1,000 test prompts

Main detector’s training sample

8,000 human posts + 470 outputs per AI model

17 × 470 = 7,990 generated examples

SP-WS standardizes selected whitespace conventions

Line breaks	Use one line-break encoding
Spaces at line ends	Remove them
Repeated blank lines	Keep one blank line
Final line break	Remove it

Punctuation and markup remain in the input

Proposed: test real document handling

Representative documents

Human and generated text from the intended setting

Real processing steps

Include ordinary editing, copying and extraction

Held-out error measurement

Report false positives and detection at a fixed cutoff